

SHORT COMMUNICATIONS

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RESTRICTED DISTRIBUTION AREA, THREAT CONDITIONS AND ADDITIONAL TWO NEW LOCALITIES OF *Bombina variegata* (L., 1758) (ANURA: BOMBINATORIDAE) IN TURKEY

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Two new localities (Hisarlı and Söğütlüdere) of the yellow-bellied toad (*Bombina variegata*) which has been recently reported for the first time in Turkey were found in the present study. We showed the restricted distribution area of the species in Turkey and tried to explain the protection and threat conditions of the yellow-bellied toad. *B. variegata* lives restricted distribution area (about 3000 hectares) in Turkey. If the species does not protect from their threats in the Turkish localities having very restricted areas, abundance of the individuals will decrease with time.

Keywords: conservation; yellow-bellied toad; habitat; Enez.

Until 2016, only one species (The red-bellied toad, *Bombina bombina*) belonging to *Bombina* genus has been known in Turkey. *B. bombina* is distributed in specific localities from Anatolian and European parts of Turkey (Yılmaz, 1986; Baran and Atatür, 1998; Baran et al., 2012). The second Bombinatorid species (The yellow-bellied toad, *Bombina variegata*) record for Turkey was done for the Edirne province (Kurtkaya locality in the vicinity of Gala Lake, Enez) by Bülbül et al. (2016).

B. variegata has a great distribution area (France in westernmost; Romania, Bulgaria, and Greece in easternmost; Germany in northernmost and Italy and Greece in southernmost) in central and southeastern Europe (Arntzen, 1978; Gollmann et al., 1997; Barbieri et al., 2004; Nürnberger et al., 2016).

The scope of the present study was: i) to add two new localities to the known distribution of *Bombina variegata* in Turkey; ii) to inform about the restricted distribution of the species; and iii) to refer to the protection and threats to species in Turkey.

During the herpetological observations in Edirne on 23 June 2016, we found two new localities for *B. variegata* in addition to Kurtkaya population recorded by Bülbül et al. (2016). The first locality is the Hisarlı Village, Enez, Edirne (GPS data: 40°43'61" N 26°13'37" E, 109 m a.s.l.). The second one is Söğütlüdere, Enez, Edirne (GPS data: 40°45'35" N 26°14'11" E, 187 m a.s.l.). The habitats of the two new localities are similar. Toads in two localities live in the small streams flowing along the valleys. A total of 17 specimens (6 ♂♂, 11 ♀♀) from Hisarlı Village and 16 specimens (6 ♂♂, 10 ♀♀) from Söğütlüdere was observed. After sexes of the toads were determined, they were released back to their natural habitats. No toads were taken as a museum example, because the number of toads was very low in both Hisarlı and Söğütlüdere populations. Only their photos were taken (Fig. 1) and GPS records were kept.

On the other hand, we visited Kurtkaya (the first locality for the *B. variegata* in Turkey) and monitored only 8 (3 ♂♂, 5 ♀♀) specimens. We visited the same locality

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Fig. 1. The lateral (A) and ventral (B) views of a female *Bombina variegata* from Söğütlüdere population.

in 2015 and the number of the toads in the population was very high when compared to our observations in 2016.

The new localities (Hisarlı and Söğütlüdere) and restricted Turkish distribution of the *Bombina variegata* are shown in Fig. 2.

Habitats: The habitats of the Hisarlı Village and Söğütlüdere generally comprise broad-leaved trees (*Quercus ilex*, *Fraxinus* sp., *Tilia* sp., *Paliurus aculeatus*, and *Acacia* sp.). Small puddles, rocks, sands and other floral elements are the main components for their habitats (Fig. 3). The localities are narrow and they are limited to living areas of about 100 – 200 m along the valleys. Other amphibians shared their living areas with *Bombina variegata* are *Pelophylax ridibundus* (Pallas, 1771), *Bufo bufo* (L., 1758), *Bufo variabilis* (Pallas, 1769), and *Lissotriton vulgaris* (L., 1758).

Conservation and Threats. Every species need to suitable habitats and habitat requirements, just like *B. variegata*. With the present study, we have shown that how much area the yellow-bellied toad is distributed in Turkey. Although *B. variegata* has a large distribution in Europe, according to our results, only about 3000 ha were used by the species in Turkey (Fig. 1). In their dis-



Fig. 2. The map showing restricted distribution area (red line) and additional two new localities (green triangles represent the localities: 1, Hisarlı Village; 2, Söğütlüdere) for *Bombina variegata* from Enez-Edirne, Turkey. The red cycle shows Kurtkaya locality in Bülbul et al. (2016) and the blue open circle represents Karacahasan Mountain.



Fig. 3. The general views of the habitats of the *Bombina variegata* from Hisarlı (A) and Söğütlüdere (B) populations.

tribution area, the toads live in the small puddles in broad-leaved forest. The slope waters that flowing from

the valley form small puddles and constitute one of the most important elements for the life of the toads. The yellow-bellied toad has been classified in LC (Least Concern) category in the red list of IUCN since 2009. However, the conservation status of the species should be handled with caution in Turkey. According to our observations, the species has lost their abundance year after year. The water supplies found in habitats of the toads have decreased. Although reproduction of the species normally has continued, maturation of the larval individuals and the number of mature individuals have decreased with time according to our observations in their habitats. The main reason of this may be pinworm drug mixtures to waters of toads by the cattle shepherd to protect the cattle from pinworm cyst infection in Kurtkaya and Söğütlüdere while the toads in Hisarlı population are highly affected by declining water supplies. The collections (for scientific or other reasons) of specimens without permission in their locality will threaten the density of populations in Turkey.

As mentioned above, yellow-bellied toad, *Bombina variegata* lives restricted distribution area (about 3000 ha) in Turkey. If the species does not protect from their threats in the Turkish localities having very restricted areas, abundance of the individuals will decrease with time.

The presence of possible new populations may be found in detailed reviews of the Karacahasan Mountain, located in the spreading area of the species in Turkey.

In conclusion, our results may be a reference for the Ministry of Forestry and Water Affairs in Turkey to start the species conservation action plan for *B. variegata* from Edirne, Turkey.

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